Python 3.12 Asynchronous Concurrency • INSTRUCTOR ANSWER KEY

Course: Python 3.12 • College: DPU • Student Name: _____

Q1. Which runtime construct is introduced in Python 3.12?

```python
import asyncio

async def fetch_service(svc_id: int) -> str:
    await asyncio.sleep(0.05)
    return f"Service {svc_id} ready"

async def main():
    async with asyncio.TaskGroup() as tg:
        t1 = tg.create_task(fetch_service(1))
        t2 = tg.create_task(fetch_service(2))
    return t1.result(), t2.result()
```

[CORRECT] A) Managed lightweight runtime scheduler

B) Static manual thread preemption

C) Unbuffered heap allocation

D) Deprecated bytecode interpreter

Key: A (Grounded in Slide 2) — Grounded in Slide 2: Structured Concurrency with TaskGroup

Distractor Analysis (Common Mistakes):